

3. (a) A student is investigating the current-voltage characteristic of a filament lamp. In the space below, **draw a circuit diagram** of the circuit she would use. [3]

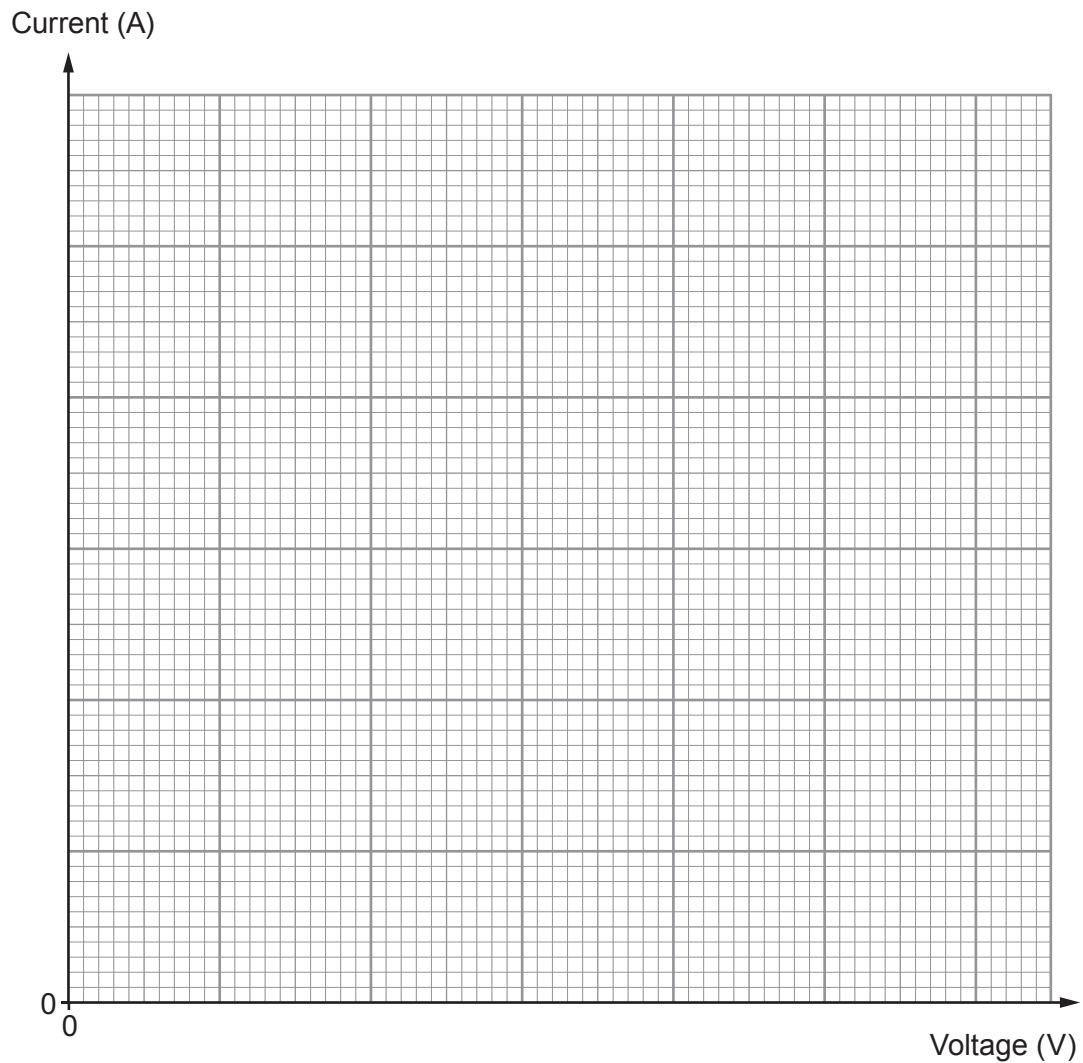
- (b) The student collects the following results.

Voltage (V)	Current (A)
0.0	0.00
2.0	0.40
4.0	0.76
6.0	0.94
10.0	1.08
12.0	1.10



- (i) Plot the data on the grid below and draw a suitable line.

[3]



- (ii) She concludes that the resistance of the lamp decreases as voltage increases. Explain using calculations, whether you agree or disagree with this conclusion. [3]

Space for calculations

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